E4-20
AC3

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2012

MSc and EEE PART IV: MEng and ACGI

CMOS INTEGRATED CIRCUIT DESIGN AND TECHNOLOGY

SAMPLE PAPER

Time allowed: 3 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s):	T.G. Constandinou
	Second Marker(s):	C. Mehring

For question 1, use the following digital model parameters given for a short-channel (50nm) CMOS technology. Note that the widths, W, and lengths, L, seen in this table are drawn lengths (minimum length is 1 while minimum width is 10)

Technology	R_n	R_p	**Scale factor**	$C_{ox} = C'_{ox} WL \cdot (scale)^2$
50 nm (short-channel)	$\frac{34k}{W}$	$\frac{68k}{W}$	50 nm	$(62.5\ aF) \cdot WL$

1. a) Discuss briefly the advantages and disadvantages of the following three logic design styles:

   (i) Static logic
   (ii) Cascode Voltage Switch Logic
   (iii) Precharge-Evaluate Logic

   [6 marks]

   b) Design in the form of transistor-level schematic diagram a complex static logic gate that implements the following Boolean function.

$$X = \left(A + B \cdot \overline{C}\right) \cdot D$$

   [6 marks]

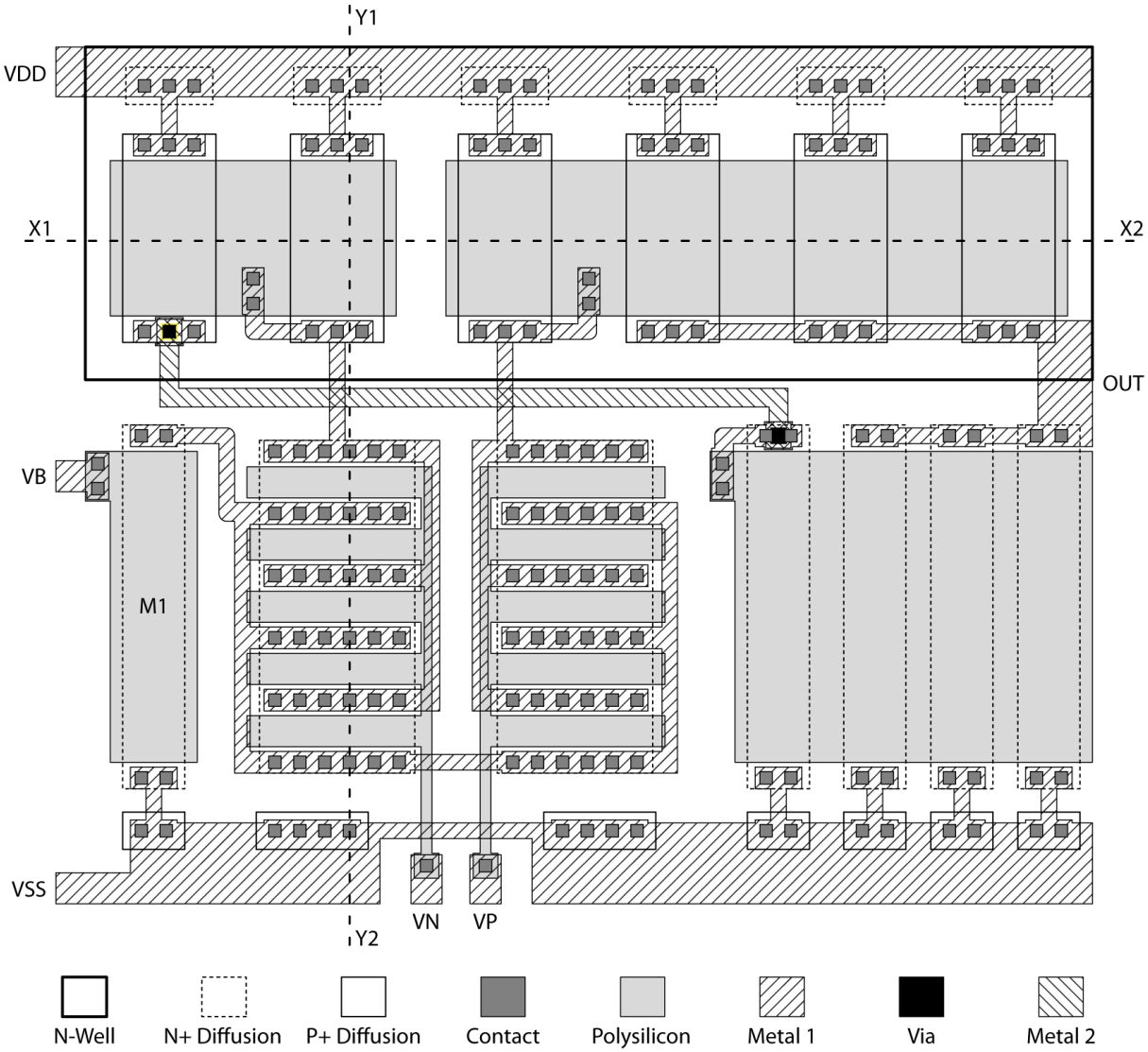
   c) Design a full custom layout for this circuit as a symbolic or stick diagram. Assume all NMOS devices are 10/1 and PMOS devices are 20/1 (drawn dimensions).

   [8 marks]

   d) Estimate the worst-case delay through the gate when driving a 50 fF load capacitance based on the device dimensions specified in part (c).

   [5 marks]

2. A full-custom layout design is shown below for an integrated circuit implemented in a commercially available 0.35µm CMOS technology.



a) Extract and draw the transistor-level circuit schematic for this design.

Given that the transistor labelled M1 is of size W/L = (2µm/10µm), label all transistors in this circuit with their sizes.

[9 marks]

b) Draw the vertical cross-sections along the lines X1-X2 and Y1-Y2. Label your diagrams indicating the n-well region and the different structures (eg. Metal1, Metal2), types and levels of doping (eg. P+, N+, etc).

[6 marks]

c) What function does this circuit perform? Determine an expression for the input/output relationship (i.e. vout/vin).

[6 marks]

d) Evaluate and comment on the quality of the layout. How could this be improved?

[4 marks]

3. *This question relates to your design project (the SAR ADC or Image Sensor):*

a) In light of the experience you gained from the design project, describe the proper procedure for designing a full-custom integrated circuit. Highlight any lessons you learned from the design project.

[6 marks]

b) Discuss the impact of *crosstalk* on your design, being a mixed-signal chip. Explain how you have engineered your solution to be robust to this through the design process.

[6 marks]

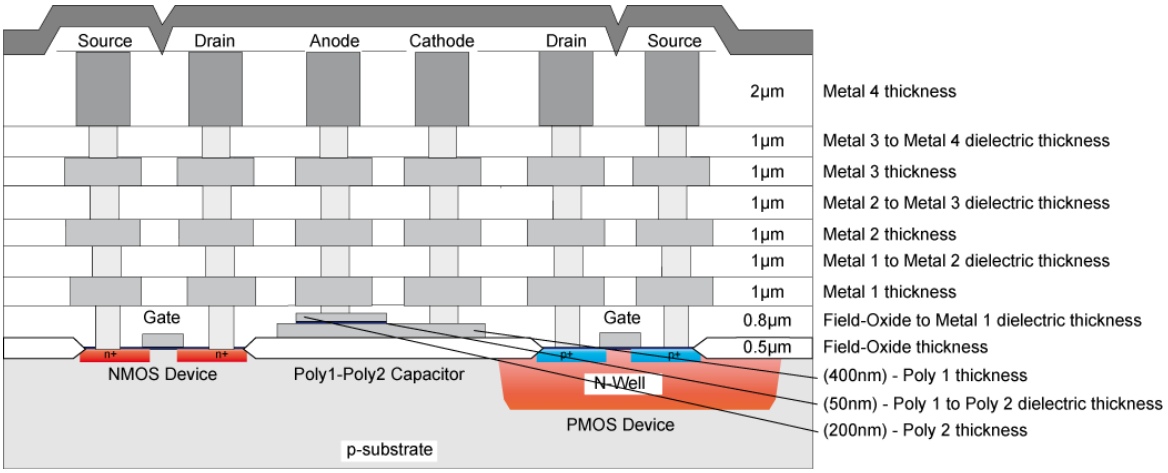
c) Sketch a timing diagram that demonstrates how your system operates. Describe which components limit its speed and how the current design can be adapted to operate at 100x its current speed (i.e. the *sampling rate* for ADC and *frame-rate* for image sensor).

[8 marks]

d) Describe your personal contributions to the project.

[5 marks]

4. Design a high-density flux capacitor in a standard 0.35µm 2P4M CMOS technology by exploiting both the vertical and lateral (side) capacitance between the interconnect layers and connecting any structure in parallel with a standard *poly1-poly2* capacitor. The relevant technology data and design rules are provided below.



Layer	**Minimum Design Width**	**Minimum Spacing**
Poly1	0.35µm	0.4µm
Poly2	0.4µm	0.45µm
Metal 1	0.5µm	0.5µm
Metal 2	0.5µm	0.5µm
Metal 3	0.5µm	0.5µm
Metal 4	0.75µm	0.65µm

a) Draw the layout and cross-section of your design for a 10µm x 10µm capacitor and identify the 2-terminals/nodes (i.e. cathode and anode).

[7 marks]

b) Calculate the total capacitance of your design in (a) and hence the capacitance per unit area (i.e. in fF/µm^2) using ε_0=8.85x10^{-15}F/m and assuming the inter-layer dielectric is SiO_2 (ε_r=4.1). Ignore the fringing-field in your calculation (i.e. consider only the vertical and lateral parallel place capacitances). What is the percentage increase in capacitance compared to using only a *poly1-poly2* capacitor?

[10 marks]

c) Explain what are the possible disadvantages of this approach?

[3 marks]

d) Describe how a MOSFET can be used as a capacitor (MOSCAP) and identify the limitations to this approach. How would you expect the capacitance per unit area to compare with a poly1-poly2 capacitor

[5 marks]

5. This question is about CMOS image sensors.

a) A typical APS-based imager uses a *rolling shutter* scheme to construct the image. Discuss the relative advantages and disadvantages between this and a *global shutter* scheme.

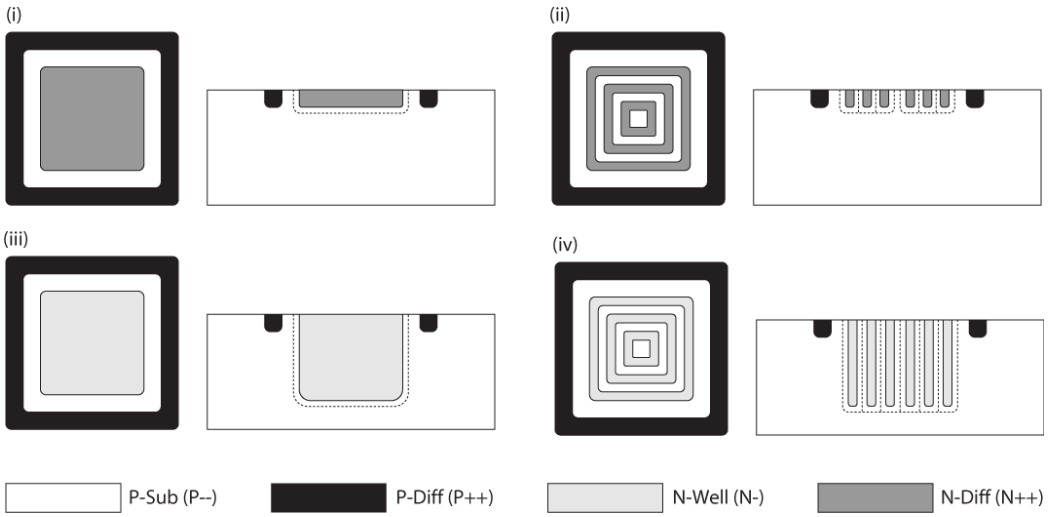
[4 marks]

b) Discuss how a global shutter could be implemented electronically and draw a possible pixel circuit schematic describing the principle of operation.

[6 marks]

c) Shown below are four different designs (layout and cross-section) of pn-junction-based photodiodes – discuss the relative performance you would expect between these.

[5 marks]



d) A photodiode can be used to generate a power supply (i.e. like a solar cell). Assuming the open-circuit voltage of a photodiode is approximately 0.5V, design a circuit to generate a stable 1.8V supply from this. *(Hint: you will need to operate the photodiode in photovoltaic mode (forward biased) to directly power an oscillator that can be used to drive a charge pump, etc).*

[10 marks]

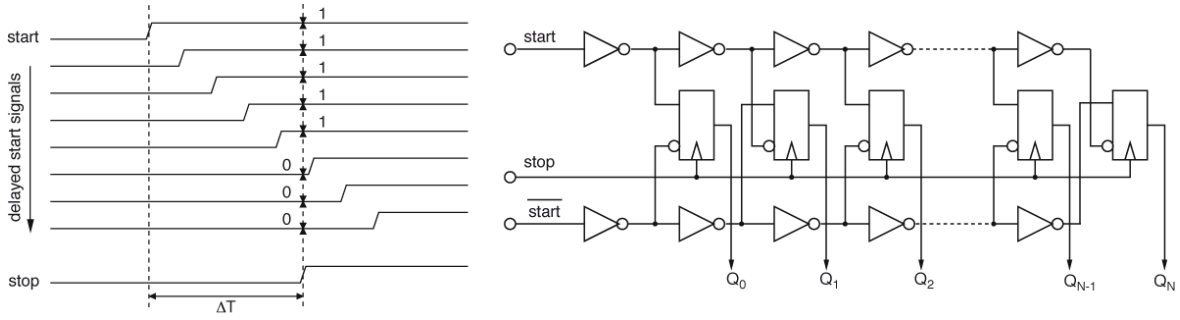
6. *Time-to-digital converters (TDC)* are used to represent an input time interval as a digital value. The most basic type of TDC is a digital counter, driven by a high-speed clock. Another implementation is based on converting the time to an analogue magnitude and then using a traditional ADC to sample the signal, similar to a slope-converter. A third possible realization is to use a cascade of inverters forming a delay line to sample the signal, to the nearest propagation delay.

   a) Discuss the relative merits and drawbacks of the three approaches of TDC described above.

   [3 marks]

   b) The principle of the inverter-based TDC is shown below. Describe how you would design a TDC based on this topology taking in a single digital input and outputting a 10-bit digital word representing the time interval the input is high (i.e. the TDC is timing the interval between the rising and falling edge). Sketch the system architecture and describe the different blocks.

   [5 marks]



   c) Using the short-channel (50nm CMOS technology) models given below, calculate the resolution (i.e time period of the LSB) of the converter and therefore the maximum input range. Assume that all NMOS devices are 5/1 and PMOS devices are 10/1 (drawn dimensions).

   [5 marks]

Technology	R_n	R_p	Scale factor	$C_{ox} = C'_{ox} WL \cdot (scale)^2$
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   d) In order to extend the input range such that the converter can cover a 24-bit resolution, it is proposed the inverter-based TDC is combined with a counter-based converter providing a coarse/fine measurement (in a similar fashion to a 2-stage flash ADC). Design the new architecture, draw the circuit schematics (either transistor or gate-level or a combination) and illustrate how it would operate by drawing a timing diagram.

   [7 marks]

   e) Discuss how your design in part (d) may be sensitive to mismatch, process variation, power supply fluctuation and/or temperature. Identify how could you improve the design to be robust to these?

   [5 marks]